

A056 v0.4.0

Circular/Fourier PKLSA–Sine-Gordon–Mittag-Leffler Falsifier

SST-Workbench

Framework v1.0.6 CANONICAL_FROZEN

1 Identification

Catalog A056, release v0.4.0. This is a new preregistered scientific release, not a threshold hotfix to v0.3.0. It replaces the raw-phase split-POD blocking test with a circular/Fourier spectral qualification while retaining the v0.3.0 metric as a non-blocking diagnostic. The release is a thin instance of SST Falsifier Framework v1.0.6 CANONICAL_FROZEN and does not vendor or patch the framework.

2 Question

For source-native E010/PKLSA-qualified vortex carriers evolved by a preregistered finite-core filament or incompressible Euler provider, does a phase-topology-aware spectral qualifier identify a stable low-dimensional Kelvin residual field, and if so do Sine–Gordon restoring dynamics and/or Mittag–Leffler fractional-memory relaxation survive blind discovery, held-out confirmation and numerical certification?

3 Hypotheses

The phase null family consists of Wave, Klein–Gordon and Duffing laws; Sine–Gordon is the phase candidate. The relaxation null family consists of exponential, stretched-exponential and bi-exponential laws; Mittag–Leffler is the fractional-memory candidate. Passing G2 is only an admissibility result. A physical conclusion additionally requires canonical-framework eligible independent/experimental replication at G7.

4 Assumptions and interpretation boundary

The provider freezes the paired Kelvin residual phase before scoring. The phase is circular, $\varphi \equiv \varphi + 2\pi$, so the v0.4.0 blocking representation uses $z = e^{i\varphi}$. The v0.3.0 $m = 3$ filament run is development/motivation evidence only and cannot close any v0.4.0 gate. Official v0.4.0 E010 campaigns use the preregistered fresh perturbation $m = 4$, $\epsilon = 0.03$. Python/NumPy FP64 is the reference lane, C++/OpenMP FP64 is certification, and SYCL FP32/DD32 remains screening-only. In conservative inviscid dynamics, projected ringdown is interpreted as transfer/dephasing unless an independent dissipative mechanism is supplied.

5 Symbols

Symbol	Meaning	Units
$\varphi(s, t)$	paired Kelvin residual phase	rad / dimensionless
$z(s, t)$	circular embedding $e^{i\varphi}$	1

$Z_n(t)$	complex spatial Fourier coefficient of z	1
P_n^W	normalized non-zero-mode power in time window W	1
$\mathcal{O}_F$	early/late Fourier-power overlap	1
η_3^{circ}	top-three circular-POD energy fraction	1
a	phase-wave coefficient	$L^2 T^{-2}$
b, c	restoring coefficients	T^{-2}
α	Mittag-Leffler fractional order	1
τ	relaxation timescale	T

6 Equations

6.1 Phase and relaxation model families

$$H_W : \varphi_{tt} = a\varphi_{ss}, \quad (1)$$

$$H_{KG} : \varphi_{tt} = a\varphi_{ss} - b\varphi, \quad (2)$$

$$H_D : \varphi_{tt} = a\varphi_{ss} - b\varphi - c\varphi^3, \quad (3)$$

$$H_{SG} : \varphi_{tt} = a\varphi_{ss} - b \sin \varphi, \quad (4)$$

$$R_{\text{exp}}(t) = e^{-t/\tau}, \quad (5)$$

$$R_{SE}(t) = e^{-(t/\tau)^\beta}, \quad (6)$$

$$R_{2e}(t) = we^{-t/\tau_1} + (1-w)e^{-t/\tau_2}, \quad (7)$$

$$R_{ML}(t) = E_{\alpha,1}[-(t/\tau)^\alpha]. \quad (8)$$

The exact null limit $E_{1,1}(-t/\tau) = e^{-t/\tau}$ remains a protected analytic test, and fractional-memory support still requires the preregistered interior interval $0.48 < \alpha < 0.97$.

6.2 Circular/Fourier G2 representation

The circular embedding and per-time spatial centering are

$$z(s, t) = e^{i\varphi(s, t)}, \quad \tilde{z}(s, t) = z(s, t) - \langle z(\cdot, t) \rangle_s. \quad (9)$$

If σ_j are singular values of the time-by-space matrix $\tilde{z}$, the top- k circular-POD energy fraction is

$$\eta_k^{\text{circ}} = \frac{\sum_{j=1}^k \sigma_j^2}{\sum_j \sigma_j^2}. \quad (10)$$

For each time sample,

$$Z_n(t) = \frac{1}{N_s} \sum_{j=0}^{N_s-1} e^{i\varphi(s_j, t)} e^{-2\pi i n j / N_s}. \quad (11)$$

For a time window W , excluding the spatial mean mode $n = 0$,

$$P_n^W = \frac{\langle |Z_n(t)|^2 \rangle_{t \in W}}{\sum_{m \neq 0} \langle |Z_m(t)|^2 \rangle_{t \in W}}, \quad n \neq 0, \quad (12)$$

and the discovery/confirmation spectral overlap is the Bhattacharyya coefficient

$$\mathcal{O}_F = \sum_{n \neq 0} \sqrt{P_n^D P_n^C}, \quad 0 \leq \mathcal{O}_F \leq 1. \quad (13)$$

The blocking G2 thresholds are preregistered as

$$\eta_3^{\text{circ}} \geq 0.95, \quad \epsilon_{\text{orth}}^{\text{circ}} \leq 10^{-10}, \quad \mathcal{O}_F \geq 0.90. \quad (14)$$

The historical raw- φ early/late POD overlap and the circular-POD early/late subspace overlap are diagnostics only; neither can block or rescue G2 in v0.4.0.

7 Algorithm

The provider resolves and re-hashes E010 source-native carriers, creates paired base/perturbed finite-core evolutions, advects material markers, and extracts $\varphi(s, t)$ and $R(t)$. G1 validates the dynamic-provider contract. G2 applies the circular/Fourier qualifier above plus analytic reference checks. Only G2-qualified cases enter the unchanged phase and relaxation model competition at G3, followed by untouched confirmation blocks at G4, deterministic resolution checks plus C++/OpenMP FP64 certification at G5, evidence-class-aware replication at G7, and mechanism classification at G8. G6 is an optional non-blocking SYCL/DD32 screening branch. The legacy raw-POD diagnostic is written to outputs for longitudinal comparison only.

Machine-readable science contract

Research question For source-native E010/PKLSA-qualified vortex carriers evolved by a pre-registered finite-core filament or incompressible Euler provider, does a phase-wrap/global-phase/spatial-shift-covariant circular/Fourier spectral qualifier identify a stable low-dimensional Kelvin residual field, and if so do Sine-Gordon restoring dynamics and/or Mittag-Leffler fractional-memory relaxation survive blind discovery, held-out confirmation and numerical certification?

Objective Replace the v0.3.0 raw-phase split-POD blocking test with a preregistered circular/Fourier G2 test that respects the phase observable topology, while retaining the historical raw-POD overlap as a diagnostic only. Then apply the unchanged SG/ML model competition and certification chain using SST Falsifier Framework v1.0.6.

H_0 H0: the invariant G2 spectral criteria fail or the Sine-Gordon/Mittag-Leffler candidates fail their preregistered discovery, held-out, numerical-certification and independent-replication thresholds beyond the registered alternatives.

H_1 H1: at least one case passes the invariant circular/Fourier G2 qualifier and one or both candidate mechanisms outperform the registered alternatives, survive held-out and numerical certification, and replicate across eligible independent physical evidence.

Registered equations

E1: Wave null phase model

$$\varphi_{tt} = a\varphi_{ss} \quad (15)$$

Dimensional check: Both sides have dimensions T^{-2} for dimensionless phase when $[a]=L^2 T^{-2}$.

E2: Klein-Gordon competitor

$$\varphi_{tt} = a\varphi_{ss} - b\varphi \quad (16)$$

Dimensional check: $[b]=T^{-2}$; every term has dimensions T^{-2} .

E3: Duffing competitor

$$\varphi_{tt} = a\varphi_{ss} - b\varphi - c\varphi^3 \quad (17)$$

Dimensional check: For dimensionless phase, $[b]=[c]=T^{-2}$.

E4: Sine-Gordon candidate

$$\varphi_{tt} = a\varphi_{ss} - b \sin \varphi \quad (18)$$

Dimensional check: $\sin(\varphi)$ is dimensionless and $[b]=T^{-2}$.

E5: Bayesian information criterion used on discovery blocks

$$\text{BIC} = n \ln(\text{RSS}/n) + k \ln n \quad (19)$$

Dimensional check: Model residuals are compared within one observable/scaling; BIC differences are dimensionless.

E6: Held-out normalized root-mean-square error

$$\text{NRMSE} = \sqrt{n^{-1} \sum_i (y_i - \hat{y}_i)^2 / \sigma_y} \quad (20)$$

Dimensional check: Numerator and denominator have the same observable units, so NRMSE is dimensionless.

E7: Mittag-Leffler function

$$E_{\alpha,\beta}(z) = \sum_{k=0}^{\infty} z^k / \Gamma(\alpha k + \beta) \quad (21)$$

Dimensional check: z must be dimensionless.

E8: Mittag-Leffler relaxation candidate

$$R_{ML}(t) = E_{\alpha,1}[-(t/\tau)^\alpha] \quad (22)$$

Dimensional check: t/tau is dimensionless.

E9: Ordinary exponential null limit

$$R_{exp}(t) = \exp(-t/\tau) = E_{1,1}(-t/\tau) \quad (23)$$

Dimensional check: t/tau is dimensionless.

E10: Stretched-exponential competitor

$$R_{SE}(t) = \exp[-(t/\tau)^\beta] \quad (24)$$

Dimensional check: t/tau is dimensionless.

E11: Bi-exponential competitor

$$R_{2e}(t) = w e^{-t/\tau_1} + (1-w) e^{-t/\tau_2} \quad (25)$$

Dimensional check: Each exponential argument is dimensionless and w is dimensionless.

E12: Phase-resolution convergence diagnostic

$$\delta_c = \|\mathbf{c}_f - \mathbf{c}_c\|_2 / \max(\|\mathbf{c}_f\|_2, 10^{-30}), \quad \mathbf{c} = (a, b) \quad (26)$$

Dimensional check: Coefficient vectors are compared componentwise under the same coordinate units; ratio is dimensionless.

E13: Memory-resolution convergence diagnostics

$$\delta_\alpha = |\alpha_f - \alpha_c|, \quad \delta_\tau = |\tau_f - \tau_c| / \max(|\tau_f|, 10^{-30}) \quad (27)$$

Dimensional check: Both diagnostics are dimensionless.

E14: Incompressible unforced Euler evolution

$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p, \quad \nabla \cdot \mathbf{u} = 0 \quad (28)$$

Dimensional check: Each acceleration term has dimensions $L T^{-2}$; incompressibility has T^{-1} .

E15: Finite-core centerline vorticity seed

$$\omega(\mathbf{x}, 0) \propto \sum_j \mathbf{T}_j \exp[-d_{per}(\mathbf{x}, \mathbf{X}_j)^2 / (2\sigma^2)] \quad (29)$$

Dimensional check: The exponential is dimensionless; normalization fixes the velocity scale.

E16: Material-marker transport

$$\dot{\mathbf{X}}(s, t) = \mathbf{u}(\mathbf{X}(s, t), t) \quad (30)$$

Dimensional check: Both sides have dimensions $L T^{-1}$.

E17: Frozen Kelvin perturbation

$$\mathbf{X}_\epsilon(s, 0) = \mathbf{X}_0(s) + \epsilon[\cos(m\theta)\mathbf{e}_1 + \sin(m\theta)\mathbf{e}_2] \quad (31)$$

Dimensional check: epsilon has dimensions L ; trigonometric factors and frame vectors are dimensionless.

E18: Complex paired transverse displacement

$$q(s, t) = \delta \mathbf{X} \cdot \mathbf{e}_1 + i \delta \mathbf{X} \cdot \mathbf{e}_2 \quad (32)$$

Dimensional check: q has dimensions L .

E19: Residual Kelvin phase

$$\varphi(s, t) = \arg[q(s, t)e^{-im\theta(s)}] \quad (33)$$

Dimensional check: The argument is an angle and therefore dimensionless.

E20: Projected mode envelope

$$R(t) = |c_m(t)|/|c_m(0)|, \quad c_m = N_s^{-1} \sum_j q(s_j, t)e^{-im\theta_j} \quad (34)$$

Dimensional check: R is dimensionless.

E21: Circular phase embedding and per-time spatial centering

$$z(s, t) = e^{i\varphi(s, t)}, \quad \tilde{z}(s, t) = z(s, t) - \langle z(\cdot, t) \rangle_s \quad (35)$$

Dimensional check: The exponential argument is dimensionless; z and $\tilde{z}$ are dimensionless complex fields.

E22: Circular-POD top-k energy fraction

$$\eta_k^{\text{circ}} = \frac{\sum_{j=1}^k \sigma_j^2}{\sum_j \sigma_j^2} \quad (36)$$

Dimensional check: A ratio of squared singular values is dimensionless.

E23: Circular Fourier/Kelvin sector amplitudes and normalized non-zero-mode power

$$Z_n(t) = N_s^{-1} \sum_j e^{i\varphi(s_j, t)} e^{-2\pi i n j / N_s}, \quad P_n^W = \frac{\langle |Z_n|^2 \rangle_{t \in W}}{\sum_{m \neq 0} \langle |Z_m|^2 \rangle_{t \in W}}, \quad n \neq 0 \quad (37)$$

Dimensional check: Z_n and normalized power P_n^W are dimensionless.

E24: Early/late Fourier-power overlap (Bhattacharyya coefficient)

$$\mathcal{O}_F = \sum_{n \neq 0} \sqrt{P_n^D P_n^C}, \quad 0 \leq \mathcal{O}_F \leq 1 \quad (38)$$

Dimensional check: The overlap is a sum over normalized dimensionless probabilities and is dimensionless.

Registered code-to-mathematics steps

1. **S1** [G0; equations E1]: Verify frozen protocol/blind commitments and enumerate hashed provider cases.
2. **S0P** [G0; equations E14, E15, E16, E17, E18, E19, E20]: Resolve and re-hash E010 v0.3.1 strict upstream-provider representatives, construct finite-core base/perturbed carriers, evolve them with the frozen provider, and commit the private carrier reveal before A056 scoring.
3. **S2** [G1; equations E1]: Validate provider metadata, array shapes, monotonicity, finite values and sampling uniformity.
4. **S3** [G2; equations E9, E21, E22, E23, E24]: Evaluate Python FP64 analytic sanity, circular-POD concentration/orthogonality, and phase-shift/spatial-shift-covariant early/late Fourier-power overlap; record raw-phase split-POD overlap as diagnostic only.
5. **S4** [G3; equations E1, E2, E3, E4, E5, E8, E9, E10, E11]: Fit Wave/Klein-Gordon/Duffing/Sine-Gordon and exponential/stretched/bi-exponential/Mittag-Leffler families on discovery blocks and evaluate BIC.
6. **S5** [G4; equations E6]: Evaluate frozen held-out temporal blocks with NRMSE ratios.
7. **S6** [G5; equations E12, E13]: Apply deterministic coarsening diagnostics and strict C++/OpenMP FP64 parity to confirmed candidates.

8. **S7** [G6; equations E12]: Consume optional canonical-framework SYCL/DD32 smoke evidence as screening-only diagnostics.
9. **S8** [G7; equations E5, E6]: Assess evidence-class-aware replication; synthetic/simulation controls cannot close physical cross-source replication.
10. **S9** [G8; equations E4, E8]: Classify joint, phase-only or memory-only physical support only after eligible replication.

8 Sources and provider contract

Real inputs satisfy **A056-DYNAMIC-PROVIDER-5**. Static knot geometry is not a substitute for $\varphi(t, s)$. **E010** supplies qualified geometry/provenance; **A056** supplies the perturbation experiment and dynamics. Each real case carries a frozen phase definition, perturbation ID, solver/provider version, carrier and upstream geometry hashes, evidence class, independence unit and provenance family. Resolution levels of one provider case are convergence replays, never independent replication.

8.1 Real-provider construction

The frozen Kelvin perturbation and observables are

$$\mathbf{X}_\epsilon(s, 0) = \mathbf{X}_0(s) + \epsilon[\cos(m\theta)\mathbf{e}_1 + \sin(m\theta)\mathbf{e}_2], \quad (39)$$

$$\dot{\mathbf{X}}(s, t) = \mathbf{u}(\mathbf{X}(s, t), t), \quad (40)$$

$$q(s, t) = \delta\mathbf{X} \cdot \mathbf{e}_1 + i\delta\mathbf{X} \cdot \mathbf{e}_2, \quad (41)$$

$$\varphi(s, t) = \arg[q(s, t)e^{-im\theta(s)}], \quad (42)$$

$$R(t) = \frac{|c_m(t)|}{|c_m(0)|}, \quad c_m = N_s^{-1} \sum_j q(s_j, t)e^{-im\theta_j}. \quad (43)$$

The volumetric lane evolves the 3-D incompressible unforced Euler equations with a 2/3-dealiased pseudo-spectral RK4 method and a Gaussian tangent-tube finite-core seed. The cheaper filament lane evolves a regularized Biot–Savart centerline directly. Both lanes are simulation evidence and cannot by themselves close physical **G7**.

9 Gates

G0 frozen protocol, score/provider hashes and blind commitments;

G1 provider/admissibility;

G2 Python-FP64 analytic plus circular-POD/Fourier-power spectral qualification;

G3 blind SG/ML discovery against registered competitors;

G4 held-out temporal confirmation;

G5 deterministic resolution checks and C++/OpenMP FP64 certification;

G6 optional SYCL FP32/DD32 screening, non-blocking for **G7**;

G7 evidence-class-aware physical cross-source replication;

G8 joint mechanism classification;

G9 commitment-verified reveal.

A **G2** failure is an admissibility failure of the preregistered spectral representation; it is not itself a Sine–Gordon or Mittag–Leffler falsification because **G3** is then not run.

10 Numerics and convergence

Phase support still requires positive Sine–Gordon coefficients, condition number $\leq 10^8$, discovery $\Delta\text{BIC} \geq 10$, and the preregistered held-out NRMSE ratio and absolute ceiling. Memory support uses the same discovery/holdout logic plus $0.48 < \alpha < 0.97$. Provider and score configuration SHA-256 values are embedded in the frozen science contract.

For $\mathbf{c} = (a, b)$,

$$\delta_c = \frac{\|\mathbf{c}_f - \mathbf{c}_c\|_2}{\max(\|\mathbf{c}_f\|_2, 10^{-30})}.$$

For memory,

$$\delta_\alpha = |\alpha_f - \alpha_c|, \quad \delta_\tau = \frac{|\tau_f - \tau_c|}{\max(|\tau_f|, 10^{-30})}.$$

Production profiles additionally require A056 C++ FP64 parity $\leq 10^{-10}$. The ringdown observable and all downstream model-selection thresholds are unchanged from v0.3.0; only the G2 spectral admission test is scientifically replaced in v0.4.0.

11 Backend authority

Lane	Precision	Authority
Python/NumPy	FP64	REFERENCE
C++17/OpenMP	FP64	CERTIFICATION
SYCL	FP32	SCREENING_ONLY
SYCL DD32 / FP32x2	two FP32 components	SCREENING_ONLY

DD32 is not IEEE FP64. The immutable SST Falsifier Framework v1.0.6 supplies the canonical bootstrap, output-integrity, reveal-orchestration, compiler/linker and backend-authority contracts. Its numerical backend implementation is unchanged from the validated v1.0.4 baseline.

12 Blindness and reveal

The framework creates nonced commitments for the private forbidden-term list and reveal payload. Blind source/output packaging excludes private/revealed content. Official v0.4.0 real-provider runs use the fresh preregistered $m = 4$, $\epsilon = 0.03$ perturbation. The observed v0.3.0 $m = 3$ cases are development-only and may be inspected only through a read-only diagnostic route. Framework mode `REVEAL_IF_ALLOWED` preserves blindness when the frozen reveal policy is not satisfied.

13 Statistics and model selection

Discovery uses

$$\text{BIC} = n \ln(\text{RSS}/n) + k \ln n.$$

Confirmation uses normalized holdout error. Sine–Gordon and Mittag–Leffler must each improve on the best registered competitor, not merely fit acceptably in isolation. The new G2 thresholds are frozen before the first official v0.4.0 $m = 4$ provider run; no threshold is tuned after observing a v0.4.0 outcome.

14 Results

No A056 v0.4.0 run summary has been generated in this report directory.

Automatic run provenance

Field	Recorded value
Protocol bundle SHA-256	4080f933a8277b548f4251f0d19b668ef3cb62c85a3480e11a627abde8b022dd
Framework schema	SST-FALSIFIER-PROTOCOL-2
Framework version	1.0.6
Python	3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0

Gate ledger

Gate	Status	Question	Reason
G0	PASS	Is the frozen A056 protocol intact and is the blind boundary verifiable?	Framework verified frozen protocol; A056 blind commitment and registered score configuration verified.
G1	PASS	Are all runtime provider cases present, hashed, contract-valid and numerically admissible?	
G2	PASS	Do Python/NumPy FP64 analytic checks and the preregistered circular-POD/Fourier-power spectral qualification pass?	
G3	FAIL	After invariant G2 qualification, does blind discovery identify preregistered Sine-Gordon and/or Mittag-Leffler candidates against the competitor families?	No candidate cleared preregistered discovery thresholds.
G4	NOT_RUN_PREREQUISITE	Do discovered candidates survive the preregistered held-out temporal confirmation blocks?	prerequisite did not PASS
G5	NOT_RUN_PREREQUISITE	Do confirmed candidates survive resolution checks and strict C++/OpenMP FP64 parity against the Python FP64 reference?	prerequisite did not PASS
G6	NOT_RUN_PREREQUISITE	If explicitly run, does optional SYCL FP32/DD32 screening satisfy the frozen framework parity contract without being promoted to certification?	prerequisite did not PASS
G7	NOT_RUN_PREREQUISITE	Does certified support replicate across eligible physical evidence classes without pseudo-replication?	prerequisite did not PASS
G8	NOT_RUN_PREREQUISITE	Does physically replicated evidence close the preregistered joint Sine-Gordon plus fractional-memory mechanism claim?	prerequisite did not PASS

G9	PASS	After the blind phase, do the reveal and blind-term commitments verify exactly?	Reveal prerequisites, blind-output integrity, and both nonced commitments verified.
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Backend summary

Field	Recorded value
A056 version	unavailable
Framework	1.0.6
Python reference	python-numpy-fp64 / unavailable / REFERENCE
C++ certification	unavailable / CERTIFICATION
Native backend	unavailable
Compiler	unavailable
Compiler version	unavailable
Build fingerprint SHA-256	unavailable
GPU screening	unavailable / SCREENING_ONLY

Source summary

Source	Role	Evidence	Provenance family
E010_V031_RELEASE	qualified_geometry_release_contract	simulation	E010_PKLSA_V031
E010_3_1_QUALIFICATION	trefoil_qualification_contract	simulation	E010_PKLSA_V031
E010_3_1_INDEPENDENCE	carrier_independence_ledger	simulation	E010_PKLSA_V031
E010_3_1_GEOMETRY_METRICS	carrier_geometry_metrics	simulation	E010_PKLSA_V031
A056_SYNTH_CONTROLS	implementation_control	synthetic_control	A056_SYNTHETIC_GENERATOR_V01

Full machine-readable provenance is preserved in `BACKEND_MANIFEST.json`, `SOURCE_MANIFEST.json`, `ENVIRONMENT_PUBLIC.json`, and `OUTPUT_MANIFEST.json`.

15 Interpretation

A pass of circular/Fourier G2 means only that the residual phase has a stable preregistered low-dimensional/cross-window spectral representation under the selected invariances. A G2 fail supports either genuine Kelvin-sector redistribution or failure of that representation; it does not by itself decide SG/ML. Synthetic-control recovery validates implementation only. E010 finite-core/Euler campaigns can establish replicated simulation conclusions but remain ineligible for physical G7. Physical support requires G5 certification plus eligible independent/experimental replication.

16 Reproducibility

The frozen protocol bundle hashes `falsifier.toml`, `science/source/gate/blind` contracts, both commitment files and this report. The science contract additionally binds the provider and scoring configuration SHA-256 values. The exact framework pin is v1.0.6 at `06_templates/SST_Falsifier_Framework/SST_Fal`. Run outputs preserve environment, source, backend, gate and output manifests; v1.0.6 verifies output-tree integrity before reveal.

References

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